30401 Mathematics and Statistics - Module 2 (Statistics) - BEMACS
FIRST PARTIAL EXAM (MOCK) - 60 minutes

Surname		Name		Student Number	

I hereby confirm my attendance at the exam.
I declare I have read the Exam Rules and I commit to respect them.
Signature:

Exercise 1 (8 points, R Dataset)

For the following questions, refer to the dataset **UN**, contained in the file: UnitedNation.RData.

The dataset contains national Statistics from the United Nations, mostly From 2009–2011 including national health, welfare, and education statistics data.
It includes 193 observations (countries) and 7 variables described below.

Region: Region of the world: Africa, Asia, Caribbean, Europe, Latin Amer, North America, Oceania.
Group: A factor with levels oecd for countries that are members of the OECD, the Organization for Economic Co-operation and Development, as of May 2012, africa for countries on the African continent, and other for all other countries. No OECD countries are located in Africa.
Fertility: Total fertility rate, number of children per woman.
GDP: Per capita gross domestic product in US dollars.
LifeExp: Female life expectancy, years.
Urban: percentage of urban population with levels high, medium, low.
InfantMortality: Infant deaths by age 1 year per 1000 live births.

a) (1pt*) Report the percentage of countries in the dataset which are from the OECD group and are in Europe.

b) (1pt*) Consider only the countries who are in the OECD group. What is the percentage of countries that are in Europe among these?

- [x] 68.4%
- [ ] 66.7%
- [ ] 13.5%
- [ ] 15.3%

c) (1pt*) Consider only the countries in the dataset with GDP per capita higher than 20,000 US dollars. Which percentage of these countries is in the OECD group?

d) (1pt**) Report the median female life expectancy of countries in Europe and the median female life expectancy of countries in Africa.

e) (1pt*) Is there an association between the variables LifeExp and Fertility? Provide a measure of linear relationship between the two variables to justify your answer, and interpret the sign and strength of the value provided.
(1pt**) Is there a causal association between the two variables? If your answer is YES, explain why. If you answer is NO, provide an example of a possible third confounding variable.

f) (1pt*) Assess the symmetry of the distribution of the variable: InfantMortality.

- [ ] It is left skewed
- [x] It is right skewed
- [ ] It is symmetric
- [ ] No clear symmetry assessment is possible

g) (1pt**) Report the names of the countries that are considered outliers with respect to infant mortality.

a) 13.47%. R Code:

```
tab=table(UN$Group,UN$Region
prop.table(tab)
```

b) 68.43%. R Code:

```
prop.table(tab,1)
```

c) 63.41%. R Code:

```
GDP20=(UN$GDP>20000)
tab=table(UN$Group,GDP20)
prop.table(tab,2)
```

d) Median female life expectancy of countries in Europe: 81.37
Median female life expectancy of countries in Africa: 58.15. R Code:

```
tapply(UN$LifeExp,UN$Region,median)
```

e) There is a strong negative association between LifeExp and Fertility, as the correlation coefficient between the two variables is: -0.8238163.
The relationship is not causal (living longer should not cause a person to have less children; having more children should not significantly shorten a person's life expectancy). A possible third confounding variable is the state of the national health service.

f) Afghanistan, Chad, Democratic Republic of the Congo, Guinea-Bissau. R Code:

```
Q1=quantile(UN$InfantMortality,0.25)
Q3=quantile(UN$InfantMortality,0.75)
IQR=Q3-Q1
UL=Q3+1.5*IQR
UN$Country[UN$InfantMortality>UL]
```

EXERCISE 2 (8 points)

With a streaming app customers can register up to 5 devices and watch content simultaneously from each of them. The table below summarises the number of devices registered for a random sample of 200 costumers.

Number of Registered Devices	1	2	3	4	5
Number of Costumers	25	85	45	25	20

a) (1pt*) Which type of data is Number of Registered Devices?

- ☐ Binary Data
- ☐ Categorical Data
- ☒ Numerical Count Data
- ☐ Numerical Continuous Data

b) (1pt*) Compute and report the five number summary (minimum, first quartile, median, third quartile, maximum) and the sample mean of Number of Registered Devices.

c) (1pt*) On the basis of b), what is your assessment of the sample distribution of Number of Registered Devices?

- ☐ It is left skewed
- ☒ It is right skewed
- ☐ It is symmetric
- ☐ No clear symmetry assessment is possible

d) (1pt*) What is the percentage of customers who registered 4 or more devices?

e) (1pt*) Assume the streaming app sets a new maximum of 4 registered devices. All the costumers in the sample who had registered 5 devices have to de-registered one and they end up with exactly 4 registered devices each.
What would happen to the sample distribution of Number of Registered Devices?

- ☐ Its mean would increase
- ☒ Its variance would decrease
- ☐ Its median would decrease
- ☐ Its third quartile would decrease

f) (3pt*) For a sample $x_1, x_2, \ldots, x_n$ of numerical data and a set of weights $w_1, w_2, \ldots, w_n$ such that $\sum_{i=1}^{n} w_i = 1$ and $w_i \geq 0$, prove that:

$$argmin_a \sum_{i=1}^{n} (x_i - a)^2 w_i = \sum_{i=1}^{n} x_i w_i.$$

(b) Min = 1 $\quad Q_1 = 2 \quad$ Median = 2 $\quad Q_3 = 3 \quad$ Max = 5

Mean = 2.65 (done in R)

(d) The required percentage is $\frac{25+20}{200} = 22.5\%$

(f) Find the minimum of the function

$$f(a) = \sum_{i=1}^{n} (x_i - a)^2 w_i$$

Calculate the first derivative:

$$\frac{\partial}{\partial a} \sum_{i=1}^{n} (x_i - a)^2 w_i = -\sum_{i=1}^{n} 2(x_i - a) w_i = f'(a)$$

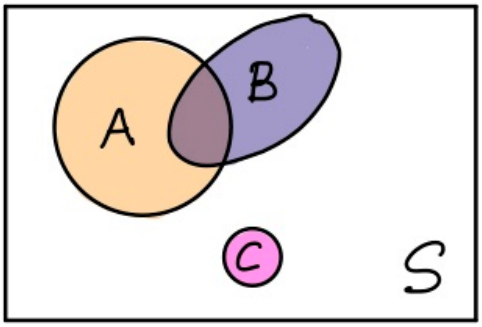
$$f'(a) = 0 \iff \cancel{-} \sum_{i=1}^{n} \cancel{2}(x_i - a) w_i = 0 \iff \boxed{\sum x_i w_i = a} \overset{=1}{\sum w_i}$$

Calculate second derivative:

$$f''(a) = +2 \sum_{i=1}^{n} w_i = +2 > 0 \implies \sum x_i w_i \text{ is the minimum}$$

EXERCISE 3 (6 points)

a) (3pt*) Let A, B and C be 3 events as represented in the Venn diagram below.



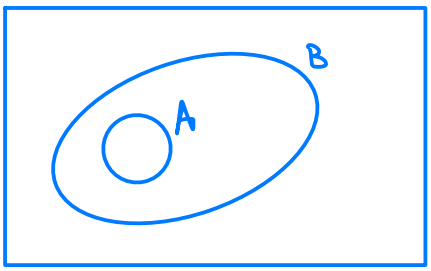
For each of the following statements decided whether it is TRUE or FALSE.

- $(A \cup B)^c = C$ [TRUE] [~~FALSE~~]
- $(A \cap B^c)^c = A^c \cup B$ [~~TRUE~~] [FALSE]
- $(C^c \cap B)^c \subseteq A$ [TRUE] [~~FALSE~~]
- $P((C^c \cap B)^c) \geq P(A)$ [TRUE] [~~FALSE~~]
- $P((C \cap (A \cap B^c))^c) = 1$ [~~TRUE~~] [FALSE]
- $P(A^c \cup C^c) = 0$ [TRUE] [~~FALSE~~]

b) (3pt**) Using the axioms of probability, prove carefully that for any (measurable) $A, B \subseteq \mathcal{S}$, if $A \subseteq B$ then $P(A) \leq P(B)$.

$= (B \cap A) \cup (B \cap A^c)$ and

$B \cap A = A$ as $A \subseteq B$.



$P(B) = P(A \cup (B \cap A^c))$

Note that $A \cap (B \cap A^c) = (A \cap B) \cap \underbrace{(A \cap A^c)}_{= \emptyset} = \emptyset$

therefore, for the <u>third axiom of probability</u> we have

$P(B) = P(A \cup (B \cap A^c)) = P(A) + \underbrace{P(B \cap A^c)}_{\geq 0} \geq P(A)$ ∎

P is a function to $[0,1]$

EXERCISE 4 (9 points)

A certain kind of valves, produced in series, have the external diameter d which has a normal distribution with mean $\mu = 12.5$ mm and standard deviation $\sigma = 0.020$ mm. According to the regulations standards the diameter should be in the interval (12.46, 12.55) (mm).

a) (1pt*) Find the probability that a valve meets the standards.

b) (1pt*) Which of the following claims from the producer is true?

- ☐ Over 99.5% of our valves do not exceed the recommended size of 12.55 mm in diameter.
- ☐ Under 1% of our valves are below the recommended size of 12.46 mm in diameter.
- ☐ Over 90% of our valves have diameter in the interval (12.47, 12.53) (mm).
- ☒ Roughly 68% of our valves are within 0.020 mm from the optimal diameter of 12.50 mm.

c) (1pt**) The base surface A of valve with diameter d is given by: $A = \left(\frac{d}{2}\right)^2 \pi$. Compute the probability that the base surface of a valve exceeds 123 mm^2.

d) (2pt***) Compute the value q such that only 1% of the valves produced have a base surface A greater than q.

should be 3 stars too

e) (2pt**) To what value should we decrease the standard deviation σ so that the probability that the random diameter d of a valve is in the interval (12.48,12.52) is exactly 95%? Assume the mean of d is still $\mu = 12.5$.

f) (2pt*) Calculate the probability to find at least 5 valves that do not meet the standards in a lot of 200. (*Hint: Define $X :=$ number of valves that **do not** meet the standards out of $n = 200$. Then approximately $X \sim Poisson(\mu')$, with $\mu' = n(1-p)$ and p is the probability that a valve meets the standards calculated in a)*)

ⓐ $d \sim N(\mu, \sigma^2)$ $\quad \mu = 12.5 \quad \sigma = 0.020$

$P(12.46 < d < 12.55) = \text{pnorm}(12.55, 12.5, 0.020) - \text{pnorm}(12.46, 12.5, 0.020)$

$= 0.97104$

ⓒ $P(A > 123) = P\left(\left(\frac{d}{2}\right)^2 \pi > 123\right) = P\left(d^2 > \frac{123 \cdot 4}{\pi}\right) =$

$= \underbrace{P\left(d < -\sqrt{\frac{123 \cdot 4}{\pi}}\right)}_{=0} + P\left(d > \sqrt{\frac{123 \cdot 4}{\pi}}\right) =$

$=0$, also diametre cannot be negative

$= 0.236836$ $\quad$ we can assume $d \geq 0$

ⓓ $P(A \leq q) = 0.99 \iff P\left(\overbrace{-\sqrt{\frac{q4}{\pi}} \leq d}\right) \leq \sqrt{\frac{q \cdot 4}{\pi}}\Big) = 0.99$

$\sqrt{\frac{q4}{\pi}} = \text{qnorm}(0.99, 12.5, 0.020) = 12.5465$

$\Rightarrow q = 12.5465^2 \cdot \frac{\pi}{4} = 123.6337$

(e) At the moment we can only solve this with trial and error.

We need σ such that

pnorm(12.52, 12.5, σ) - pnorm(12.48, 12.5, σ) = 0.95

define a vector:

sigma = seq(0.01, 0.02, 0.001) and evaluate the probability above.

p = pnorm(12.52, 12.5, sigma) - pnorm(12.48, 12.5, sigma)

(you can use cbind(sigma, p) to spot the sigma that grants the probability closest to 0.95)

a very good estimate seems $\sigma = 0.0102$

(f) $X \sim \text{Poisson}(\mu')$ $\quad \mu' = n(1-p)$ $\quad n = 200$ $\quad p = 0.9710$

$P(X \geq 5) = 1 - P(X \leq 4) = 1 - \text{ppois}(4, 5.79196) = 0.6861325$